SCIENCE · CLASS 8

Friction

Name: _____

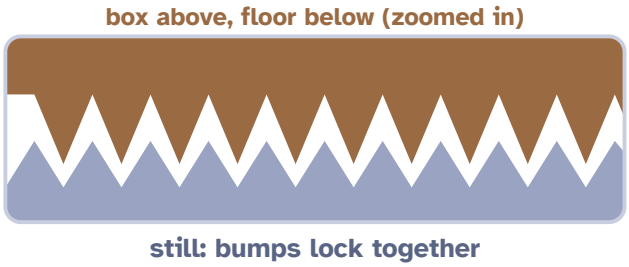
Notes · what friction is, why the first push is hardest, and how we change it

Date: _____

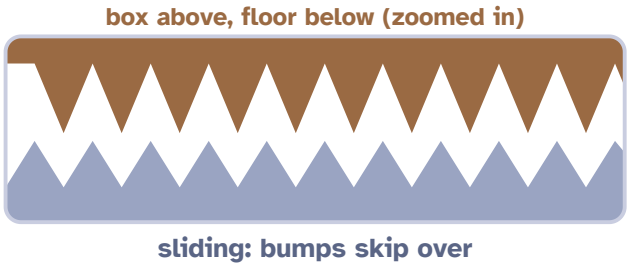
1. What is friction?

Friction is a force that **opposes motion** (or an attempt to move) between two surfaces that touch each other. It acts along the surfaces, opposite to the direction of motion.

Why does it happen? Even smooth-looking surfaces have tiny bumps. When two surfaces press together, the bumps **lock into each other**.



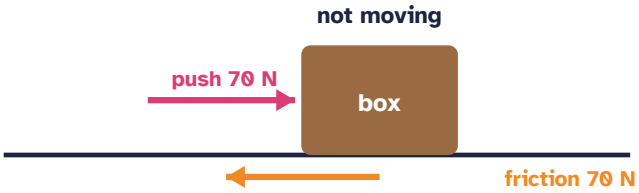
still: bumps lock together

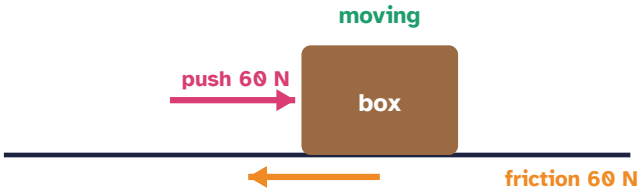


sliding: bumps skip over

2. Why is the first push the hardest?

The friction that stops a still object from moving (**static friction**) is **more** than the friction while it slides (**sliding friction**). When still, the bumps have time to lock firmly; when sliding, they don't.





Example A 20 kg box needs **100 N** to start moving but only **60 N** to keep it moving at a steady speed.

Remember **static friction > sliding friction > rolling friction**. That's why wheels make moving things easy.

3. What decides how much friction there is?

- **The type of surfaces:** rough surfaces (sandpaper, gravel) have more friction than smooth ones (glass, ice).
- **How hard they are pressed together:** a heavier box has more friction.

4. Friction: a necessary evil

Useful: we need it to

- walk without slipping
- hold a pen and write
- stop a bicycle with brakes
- light a matchstick

Harmful: it also

- wears out shoe soles and tyres
- heats up machine parts
- wastes energy in engines

5. Increasing and reducing friction

To increase friction	To reduce friction
treads on tyres and shoe soles	oil or grease (lubricants)
gymnasts' chalk powder on hands	ball bearings in fans and cycles
sand on a slippery road	wheels on suitcases

Fluid friction (drag): air and water also oppose motion. Fish, birds and aeroplanes have **streamlined** shapes to reduce it.

Common mistake Saying friction always "stops" things. It only opposes **relative** motion. Without friction, your feet couldn't push the ground to walk!

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Friction: worksheet

Name: _____

Try every question. Show your working.

Date: _____

A. Choose the correct answer

1. Friction acts (a) along the motion (b) opposite to the motion (c) upwards
2. Which is the smallest? (a) static friction (b) sliding friction (c) rolling friction
3. Gymnasts rub chalk powder on their hands to (a) increase friction (b) reduce friction
4. Ball bearings are used to (a) increase friction (b) reduce friction

B. Push the box

A box needs **more than 80 N** to start moving. Once it slides, friction is **50 N**.

5. You push with **70 N**. Does the box move? How big is the friction?
6. You push with **90 N**. Does it move? Once it slides, what is the net force on it?
7. How hard must you push to keep it sliding at a **steady speed**?

C. Increase or reduce friction?

8. deep treads on tyres: _____
9. oiling a squeaky door hinge: _____
10. sand on an icy road: _____
11. wheels on a suitcase: _____
12. spikes on running shoes: _____
13. polishing a carrom board: _____

D. Think

14. Why does a matchstick catch fire when you strike it on the box?

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Friction: answer key

With short working · every number checked by code

A. Choose

1. (b) opposite to the motion **2.** (c) rolling friction **3.** (a) increase **4.** (b) reduce

B. Push the box

5. 70 N is not more than 80 N, so the box **does not move**. Friction matches your push: **70 N**.

6. 90 N > 80 N, so it **starts to move**. While sliding: net force = 90 − 50 = **40 N** (so it speeds up).

7. Push = sliding friction = **50 N**. Then the net force is 0 and the speed stays the same.

C. Increase or reduce

8. increase **9.** reduce **10.** increase **11.** reduce **12.** increase **13.** reduce

D. Think

14. Friction between the match head and the rough side of the box produces heat. The heat raises the temperature of the chemicals on the match head until they catch fire.

Original questions by Adithya Vikram · matches the Class 8 Short "Why is the first push the hardest?" · every numerical answer computed and double-checked by code.